

SSV Conclusions A

Finite update rate and presentist emergent gravity:
what the programme leaves to mainstream physics

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Closing note, 2026-06-16

Abstract

The Saturated Superfluid Vacuum (SSV) programme is closed: no surviving route derives the quantities it set out to derive. This note records what of its *ontology* maps cleanly onto standing physics, so the conceptual content is not lost with the framework. The central picture — local time as the rate of phase update of an order parameter, a finite update capacity that loading consumes, and gravity as the response of a holographic record to that loading — is a coherent position, but it is not new and it makes no prediction beyond the programmes it restates. It is the entropic/thermodynamic emergent-gravity programme (Jacobson, Verlinde, Padmanabhan) on a superfluid vacuum (Volovik, Zloschastiev), with time read as a Josephson phase rate and a presentist reading of the emergent metric. Its honest home is foundations / philosophy of physics: interpretive coherence, not a gravity theory. Following the repository’s standing rule, the magnitude failures are stated at full strength and not softened.

1 Purpose and the honest frame

This is a conclusions document, not a research paper. It exists because the SSV ontology — specifically *time delay from limited update capacity* and *gravitational bending as a holographic-screen update* — is the part of the programme that still feels like it might say something, and the owner asked whether it can be adapted to mainstream physics. The answer below is: the ontology is real, recognisable, and already named in the literature; what it adds on top is interpretation rather than prediction; and the reasons it cannot be more are the same reasons the programme closed. Per the repository’s negative results rule, those reasons are stated plainly.

2 The ontology in one paragraph

Physical time is identified with the local rate of change of the vacuum order parameter $\Psi = \sqrt{\rho} e^{i\theta}$. In a condensate the phase obeys a Josephson relation,

$$\hbar \dot{\theta} = -\mu, \tag{1}$$

so the local “tick” is set by the local chemical potential. The medium has a *finite* capacity to update its state; loading it (placing energy/defects in it) consumes update capacity and slows the local tick. Where the slowing is uneven it has a gradient, and any motion through the gradient is bent toward the region of greatest delay. That bending is gravity (Paper IV). In

the holographic reading (Paper VII-b, the “Option D” fork) nothing bends space in the bulk: Ψ lives *on* space, the loading is written to a self-defined acoustic horizon as an area-law record, and the read-back of that record steers trajectories. Matter writes the ledger; the ledger dents capacity; the dent is gravity.

3 Where each component already lives

- **Time as phase rate $\rightarrow$ Josephson + analogue g_{00} .** Promoting $\hbar\dot{\theta} = -\mu$ to an emergent lapse is standard analogue gravity: the acoustic metric’s g_{00} is fixed by the local condensate state (Unruh 1981; Visser 1998; Volovik). Weak-field, this reproduces $g_{00} = 1 + 2\Phi/c^2$ and the Einstein-1911 (time-delay) half of light deflection. This is exposition of a known result.
- **Finite update capacity $\rightarrow$ Margolus–Levitin / Lloyd.** The one genuinely distinct gloss — a finite *rate* of state change — has a rigorous mainstream anchor: the maximum rate of orthogonal-state evolution is bounded by energy, $\nu_{\perp} \leq 2E/\pi\hbar$ (Margolus–Levitin 1998; Lloyd, *Ultimate physical limits to computation*, 2000). SSV’s “update capacity” is a physical-substrate gloss on this bound. It does not, however, turn the bound into a new gravitational prediction.
- **Time as change, not backdrop $\rightarrow$ relational time.** The claim that time is the medium changing, not a dimension it moves through, is the relational tradition: Barbour’s time-less mechanics and the Connes–Rovelli thermal time hypothesis (1994). SSV is a concrete realisation, not a new argument.
- **Screen update as gravity $\rightarrow$ Jacobson + Verlinde + Padmanabhan.** “Gravity is the holographic screen updating” is, almost verbatim, entropic / thermodynamic gravity: Jacobson 1995 (the Einstein equation as an equation of state, $\delta Q = T dS$), Verlinde 2011 (gravity as an entropic force from holographic screens), and Padmanabhan’s gravity-as-thermodynamics. The Option D arc runs through exactly this machinery.

4 What the ontology adds, and what it does not

What it adds is *interpretive coherence*: a single medium in which the same phase field carries the local clock (rate $\dot{\theta}$), the static near-field (gradient $\nabla\theta$), and the long-range encoding (the gapless propagating mode). That unification is genuinely attractive as a picture.

What it does not add is a falsifiable prediction. The construction reproduces the *form* of the weak-field metric, the area law, and the Unruh/Hawking temperature, and concedes every *magnitude*: Newton’s constant G , the cosmological scale Λ/H_0 , and the MOND-like acceleration a_0 all enter as inputs rather than outputs. “Form yes, magnitude no” is not an incidental gap; it is the wall the programme hit. Conclusions B records the sharpest instance of it.

5 An internal tension the salvage must own

The two halves of the ontology are, on inspection, two different ontologies:

- *Update capacity (Paper IV)* is **bulk-realist**: the medium’s local state physically slows time and the gradient bends motion.

- *Holographic screen (Option D)* is **anti-bulk-realist**: “nothing bends space in the bulk; space was never bulk-dynamical; geometry is localised into the encoding.”

These reconcile only through a **duality** claim — bulk and screen are two complete descriptions of one state, with no independent dials. That duality is the most rhetorically powerful element of the picture and the one with the least empirical content: by the programme’s own audit it “carries no falsifier of its own.” Any honest adaptation should present the duality as a heuristic frame (the Wheeler/Deser “laws as closure conditions” lineage), not as a mechanism.

6 Presentism: the one place the ontology pulls its weight

SSV is natively presentist. There is local becoming — the update tick, a privileged substrate “now” — but no completed, global, infinite time and no block universe. This is legitimate precisely *because* SSV’s spacetime is emergent rather than fundamental: an effective metric with a domain of validity is under no obligation to be geodesically complete, so there is no real future infinity $\mathcal{I}^+$ for a global construction to require. This is not idle: it dissolves the obstacle that breaks global de Sitter holography (the non-unitary Euclidean CFT at $\mathcal{I}^+$ has no substrate referent) and routes naturally to *observer / static-patch* holography, where well-defined observables and a finite entropy require including the observer’s worldline — the “now” — as in the de Sitter observer-algebra programme (Chandrasekaran–Longo–Penington–Witten 2022). The caveat, stated plainly: the present cosmological horizon still has a Gibbons–Hawking entropy whose microstates SSV does not supply, so presentism removes an obstacle without building the dual.

7 Verdict

The ontology — finite-update-rate time dilation plus holographic-screen gravity — is a coherent foundations position that restates the entropic/emergent-gravity programme on a presentist superfluid vacuum. It is adaptable to mainstream physics as **philosophy of physics**: a careful essay on time as finite update rate and on presentist emergent gravity, engaging Jacobson, Verlinde, Padmanabhan, Barbour, Connes–Rovelli, and the de Sitter observer-algebra work. It is *not* adaptable as a new theory of gravity: it predicts nothing those programmes do not, it inherits their open empirical objections (e.g. the qBounce and interferometric constraints on entropic gravity), and in concrete form it fails on magnitude.

Do not reclaim under a new banner. The closed no-gos remain closed: deriving G , a spin-2 graviton or tensor gravitational-wave polarisations, a fresh resolution of Loschmidt’s paradox, and a_0 as a selected prediction. These are recorded as falsified or conceded and should not be re-imported into a foundations write-up as live claims.

References

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